

CIE $L^*a^*b^*$ space

L^* indicates lightness and a^* and b^* are chromaticity coordinates.

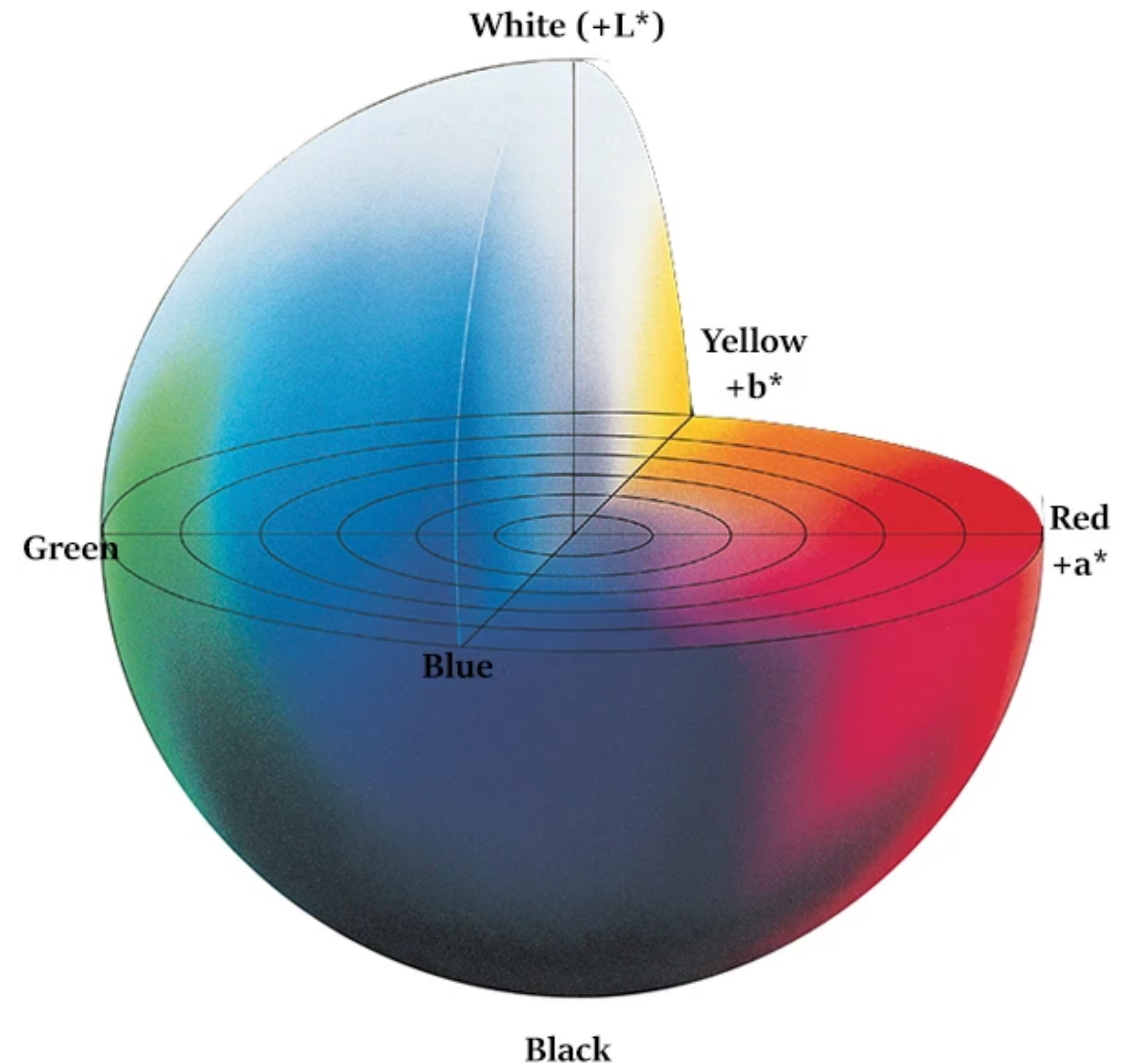
$+a^*$ is the red axis

$-a^*$ is the green axis

$+b^*$ is the yellow axis

$-b^*$ is the blue axis

Perceptually uniform: a nonlinear transformation of XYZ



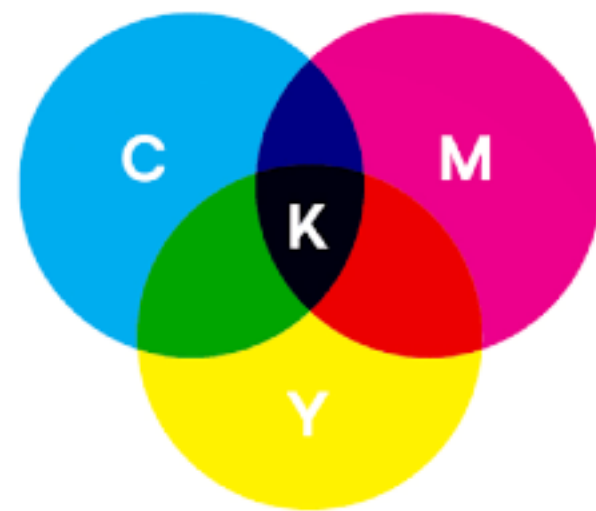
RGB color space

Each screen pixel is driven by three small and very close but still separated light sources.

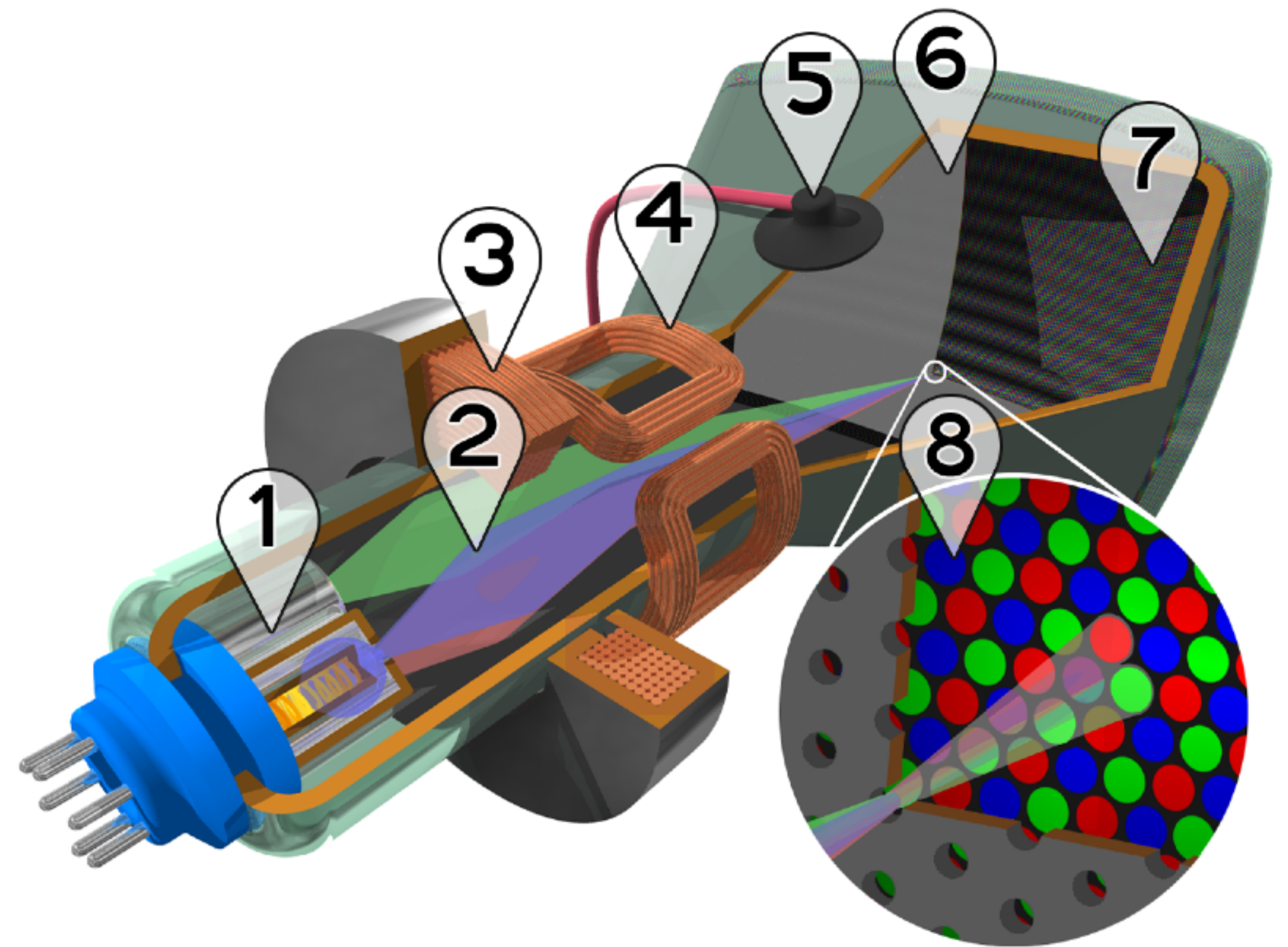
At common viewing distance, the separate sources are indistinguishable, we interpret them as a solid color.



additive (light)



subtractive (pigment, ink)



CRT monitor